

**THE THIRTY-EIGHTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

December 3, 1977

Problem A-1

Consider all lines which meet the graph of

$$y = 2x^4 + 7x^3 + 3x - 5$$

in four distinct points, say (x_i, y_i) , $i = 1, 2, 3, 4$. Show that

$$\frac{x_1 + x_2 + x_3 + x_4}{4}$$

is independent of the line and find its value.

Problem A-2

Determine all solutions in real numbers x, y, z, w of the system

$$x + y + z = w,$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{w}.$$

Problem A-3

Let u, f , and g be functions, defined for all real numbers x , such that

$$\frac{u(x+1) + u(x-1)}{2} = f(x) \quad \text{and} \quad \frac{u(x+4) + u(x-4)}{2} = g(x).$$

Determine $u(x)$ in terms of f and g .

Problem A-4

For $0 < x < 1$, express

$$\sum_{n=0}^{\infty} \frac{x^{2^n}}{1 - x^{2^{n+1}}}$$

as a rational function of x .

Problem A-5

Prove that

$$\binom{pa}{pb} \equiv \binom{a}{b} \pmod{p}$$

for all integers p, a , and b with p a prime, $p > 0$, and $a \geq b \geq 0$.

Notation: $\binom{m}{n}$ denotes the binomial coefficient $\frac{m!}{n!(m-n)!}$.

Problem A-6

Let $f(x, y)$ be a continuous function on the square

$$S = \{(x, y) : 0 < x < 1, 0 < y < 1\}.$$

For each point (a, b) in the interior of S , let $S_{(a,b)}$ be the largest square that is contained in S , is centered at (a, b) , and has sides parallel to those of S . If the double integral $\iint f(x, y) dx dy$ is zero when taken over each square

$S_{(a,b)}$, must $f(x,y)$ be identically zero on S ?

Problem B-1

Evaluate the infinite product

$$\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1}.$$

Problem B-2

Given a convex quadrilateral $ABCD$ and a point O not in the plane of $ABCD$, locate point A' on line OA , point B' on line OB , point C' on line OC , and point D' on line OD so that $A'B'C'D'$ is a parallelogram.

Problem B-3

An (ordered) triple (x_1, x_2, x_3) of positive *irrational* numbers with $x_1 + x_2 + x_3 = 1$ is called "balanced" if each $x_i < 1/2$. If a triple is not balanced, say if $x_j > 1/2$, one performs the following "balancing act"

$$B(x_1, x_2, x_3) = (x'_1, x'_2, x'_3),$$

where $x'_i = 2x_i$ if $i \neq j$ and $x'_j = 2x_j - 1$. If the new triple is not balanced, one performs the balancing act on it. Does continuation of this process always lead to a balanced triple after a finite number of performances of the balancing act?

Problem B-4

Let C be a continuous closed curve in the plane which does not cross itself and let Q be a point inside C . Show that there exist points P_1 and P_2 on C such that Q is the midpoint of the line segment P_1P_2 .

Problem B-5

Suppose that $a_1, a_2, \dots, a_n$ are real ($n > 1$) and

$$A + \sum_{i=1}^n a_i^2 < \frac{1}{n-1} \left(\sum_{i=1}^n a_i \right)^2.$$

Prove that $A < 2a_i a_j$ for $1 < i < j < n$.

Problem B-6

Let H be a subgroup with h elements in a group G . Suppose that G has an element a such that for all x in H , $(xa)^3 = 1$, the identity. In G , let P be the subset of all products $x_1 a x_2 a \cdots x_n a$, with n a positive integer and the x_i in H .

- Show that P is a finite set.
- Show that, in fact, P has no more than $3h^2$ elements.